Both credit and signature-based debit card transactions are typically processed in batch mode at the POS, and settlement is delayed until the batches are processed at the end of the day. PIN-based debit card transactions, although processed in real time at the POS, typically settle at the end of the day using the ACH. Merchants often prefer that customers use PIN-based debit cards due to the lower costs associated with these transactions over the costs for signature-based credit and debit cards. With PIN-based transactions, the consumer must apply the pre-established PIN to validate the transaction. Each of these types of card payments is described below.

In the United States, almost all cards are magnetic-strip-based, while in Europe and Asia, consumer account information is often stored on a computer chip embedded in the card. These computer-chip-based systems have more security features than the magnetic strip systems; therefore, more financial institutions and merchants in the U.S. are adopting chip processing infrastructure. Consumers have welcomed recent initiatives with chip-based contactless cards so, the growth in these chip-based-cards is expected to continue.

In general, credit cards have revolving credit arrangements that allow consumers to make purchases and be billed later. Most credit card accounts allow the consumer to carry a balance from one billing cycle to the next and make a minimum payment in each billing cycle (e.g., two to three percent of their total balance) rather than requiring payment of the full balance.

A charge card is a specific kind of credit card that has a short-term, fixed-period credit arrangement. The balance on a charge card account is payable in full when the statement is received and cannot be rolled over from one billing cycle to the next. This arrangement exposes the issuing institution to less credit risk than open-ended accounts.

Financial institutions are important participants in various credit card systems. They issue and distribute cards, clear and settle the associated payments, and act as, or sponsor, merchant acquirers. [30] There is an increasing concentration of both credit card issuers and processors within the marketplace as larger issuers are bringing processing functions in-house. Some large institutions have exited the credit card issuance and processing businesses due to lack of economies of scale.

This booklet groups credit or charge cards in three categories: general-purpose credit cards, co-branded/affinity cards, and private label (store) cards.

General Purpose Credit Cards

General-purpose cards have the logo of one of the bankcard companies on the front. [31]

These cards are associated with the consumer's or cardholder's revolving credit account at a financial institution or other business. The revolving credit line is capped or limited based on the creditworthiness of the consumer. These cards can be used at any location that accepts credit cards from the particular bankcard company and include bankcards and closed-loop cards. Bankcards require agreements and transaction processing arrangements among participants, while closed-loop cards may not.

- Financial institutions issue bankcards in conjunction with the three major credit card association networks, Visa, MasterCard, and American Express. MasterCard, Visa, and American Express operate "open" networks in which financial institutions can compete in card-issuing and merchant acquiring. The card-issuing financial institution and the merchant acquirer can be different organizations. Firms that serve as both the card issuing agent and the merchant acquirer issue closed loop credit cards.

Co-Branded/Affinity Credit Cards

Some merchants and organizations form marketing arrangements with financial institutions to issue general-purpose credit cards with the merchant or organization name on the front of the card. These cards are termed co-branded or affinity cards and the card accounts may be part of the bankcard company networks.

Co-branded cards typically offer consumers a rewards program. Organizations such as sports teams, schools, or service organizations issue affinity cards jointly with a financial institution that offers compensation in return for marketing to the merchant's customers or the organization's members. The institution might base its compensation on the number of account applications, the number of accounts activated, account volume and income, or other defined benchmarks.

Private Label (Store) Credit Cards

In some cases, financial institutions might issue a card jointly with a merchant. These cards are known as private label or store cards. Consumers can use them only at the merchant whose name appears on the front of the card. These cards do not carry a bankcard company logo, and the merchant typically plays a limited role in the issuance of the card or managing the credit relationship. [32]

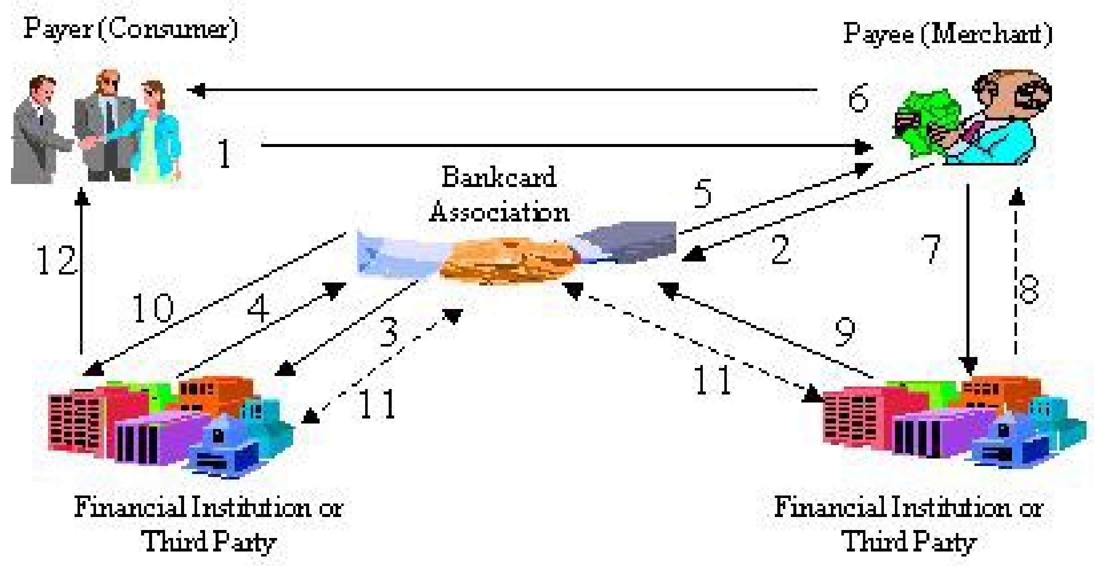
Bankcard Companies

The two major bankcard companies, Visa and MasterCard, account for the majority of credit and debit cards in use. Both organizations began as bank service companies, owned by principal-member financial institutions. They provide separate, but similar operating policies, procedures, and controls for bankcard issuance, acquiring, and settlement activities. The companies own the credit card trademark, granting membership to financially sound financial institutions that apply. Only members are allowed to issue cards bearing the company logo, and they pay transaction and membership fees for use of the bankcard association logo and services.

Each company has three primary types of membership: Visa has principal, associate, and participant memberships; MasterCard has principal, affiliate, and agent memberships. Each membership type conveys different privileges. Principal membership allows members to solicit cardholders and issue cards, solicit and sign merchants, and sponsor other financial institutions for membership in the company. Associate/affiliate and participant/agent members can perform all of the principal membership functions except sponsor other members.

Card issuers are financial institutions that have permission to issue bankcard company credit cards. Acquiring financial institutions and sponsored third parties have contracts with merchants that accept a bankcard company's products. Acquiring financial institutions accept and process transactions from those merchants through the company's network interchange payment system. The cost of technology infrastructure and the level of transaction volume are high for bankcard-acquiring institutions. Most rely on third-party service providers. [33] Under the bankcard company's bylaws, acquiring financial institutions are responsible for the actions of all contracted third-party service providers; therefore, they are expected to monitor carefully the providers' compliance with the companies' operating rules.

The bankcard companies set interchange fees, which are paid by the merchant acquirer to the issuing financial institution. The merchant acquirer typically passes this fee along with a discount or acquirer fee for processing services to its merchants. Bankcard issuing institutions generate their revenue from the interest charged on revolving balances, and from the interchange, late, over-limit, cash advance, and card fees. Merchant-acquiring institutions, which assist in clearing and settling credit card transactions, generate most of their revenue from the acquiring and other processing fees (e.g., charge-back processing and account maintenance) they charge to the merchant.



Legend: Solid lines represent the flow of information and dashed lines represent the flow of funds.

Figure 5: Credit Card Clearing and Settlement

Figure 5 illustrates the payment and information flows for a typical credit card transaction. In this example, the consumer pays a merchant with a credit card (step 1). The merchant electronically transmits the data, at the POS and through the bankcard company's electronic network, to the card issuer for authorization (steps 2 and 3). If approved, the merchant receives the authorization to capture funds, and the cardholder accepts liability by signing the credit voucher (steps 4, 5, and 6). In cases involving purchases under $25, the cardholder does not have to sign. The merchant receives payment, net of fees, by submitting captured credit card transactions to its financial institution in batches or at the end of the day (steps 7 and 8). The merchant acquirer forwards the sales draft data to the bankcard company, who forwards the data to the card issuer (steps 9 and 10). The bankcard company determines each financial institution's net debit position. The bankcard company's settlement financial institution coordinates issuing and acquiring settlement positions. Members with net debit positions (generally issuers) send owed funds to the company's settlement financial institution, which transmits owed funds to the merchant acquirers. The settlement process takes place using a separate payment network such as Fedwire® (step 11). [34] The card issuer will then present the transaction on the cardholder's next monthly statement (step 12). The cardholder makes a payment for the charges incurred in accordance with the cardholder agreement.

Debit and ATM Cards

Debit cards are associated with an existing transaction account at a financial institution. The card enables consumers to access their accounts for a variety of transactions. Debit cards are either online (i.e., PIN-based) or off-line (i.e., signature-based).

- Online (PIN-based) debit cards have been available for several decades and have seen significant growth since the early 1990's. Online debit cards use a PIN for customer authentication and online access to account balance information. At present, financial institutions authenticate customers by matching the PIN with the account number directly through a merchant's terminal. Debit card transactions are authorized in real time at the POS using the same electronic funds transfer (EFT) networks that handle ATM transactions and are typically settled at the end of the day using the ACH network. Customers may also receive cash at the POS because messaging between the financial institution and the retailer confirms funds availability. Merchants prefer PIN-initiated card transactions as the processing fees are substantially lower. Also, credit risk is shifted to the customer as the merchant's responsibility for authentication is greatly reduced.
- Off-line (signature-based) debit cards were introduced in the late 1980's by Visa and MasterCard. Consumers are using them increasingly at merchant locations that accept bankcards. Off-line debit card systems authenticate consumers through a written signature or other authenticating action. The transactions are processed in batch mode through the same bankcard networks as credit card transactions and typically settle at the end of the business day. Generally a cardholder can use an off-line debit card anywhere that accepts a similar online transaction.

The use of biometric technology as a means to authenticate payments is also growing because of its convenience and perceived security features. Available technologies allow customers to pay for purchases by placing a finger on a sensor, which links the image to the customer's account using a simple method of finger scanning at check out. Societal implications and security concerns surrounding the use of biometric identification may act as impediments to market acceptance.

Financial institutions issue ATM cards to consumers to provide online access to account information and to allow consumers to make withdrawals and deposits at ATMs.

Consumers typically enter a PIN for authentication at an ATM, although other authentication methods such as biometric technology are available. Consumers may use an ATM deployed by other financial institutions or third parties but typically will pay fees to the ATM owner and their own financial institution. Many financial institutions now offer ATM cards that can also be used as debit cards for POS transactions at participating merchants.

Decoupled Debit Cards

Decoupled debit cards permit a financial institution to issue a debit card to consumers regardless of where their demand deposits or other transaction accounts are held. The term "decoupled" is derived from the separation of the traditional relationship between the debit card issuer and the financial institution that provides the transaction deposit account. The decoupled debit card transaction between the consumer and merchant is processed through one of the card-branded networks or an alternative proprietary network. Instead of using the EFT networks used for debit card products, the issuer uses the ACH network to debit the consumer's account for settlement.

By decoupling the debit transaction from the bank where the consumer has the depository relationship, the intermediary can capture the interchange revenue from the card transaction. A part of this product's initial appeal was the cost efficiency derived from bundling transactions prior to entry into the ACH network for settlement. However, a recent NACHA Rule Interpretation issued on November 9, 2007 [35] prohibits the aggregation of individual debit transactions prior to settlement through the ACH, and instead requires the issuer to pay ACH origination fees on each discrete transaction conducted during the course of a day. The interpretation was issued in response to concerns that bundling transactions through the ACH might mask risks that are transparent in individual transactions and unintentionally subvert risk management tools used by financial institutions that receive payment through the ACH. Decoupled debit card programs that rely on transaction bundling may need to be re-engineered to comply with the new interpretation.

The risk profile for decoupled debit card issuers differs from a debit card program because payments are settled through the ACH, creating a delay from the time the card transaction is initiated and exposing the issuer to credit risk. With a traditional debit card, a financial institution can verify the availability of funds before the transaction is authorized. With decoupled debit transactions, credit risk exposure may arise from faulty account verification or insufficient deposit account balances. Financial institutions that issue decoupled debit products should implement risk management programs to mitigate and control these new risks associated with the nontraditional customer relationship.

EFT/POS Networks

EFT/POS networks process, route, clear, and settle ATM and online POS debit card transactions by linking financial institution card issuers and merchant acquirers, consumers, merchants, and third-party service providers through telecommunication gateways. The primary functions of the networks include routing transactions through central switching gateways, acting as clearing houses to settle network member on-us transactions, and forwarding "foreign" nonmember transactions for processing. Both credit card and signature-based debit card transactions are processed in batch mode at the POS, and settlement is delayed until the batches are processed at the end of the day. PIN-based debit card transactions typically settle at the end of the day using the ACH, although they are authorized in real time at the POS.

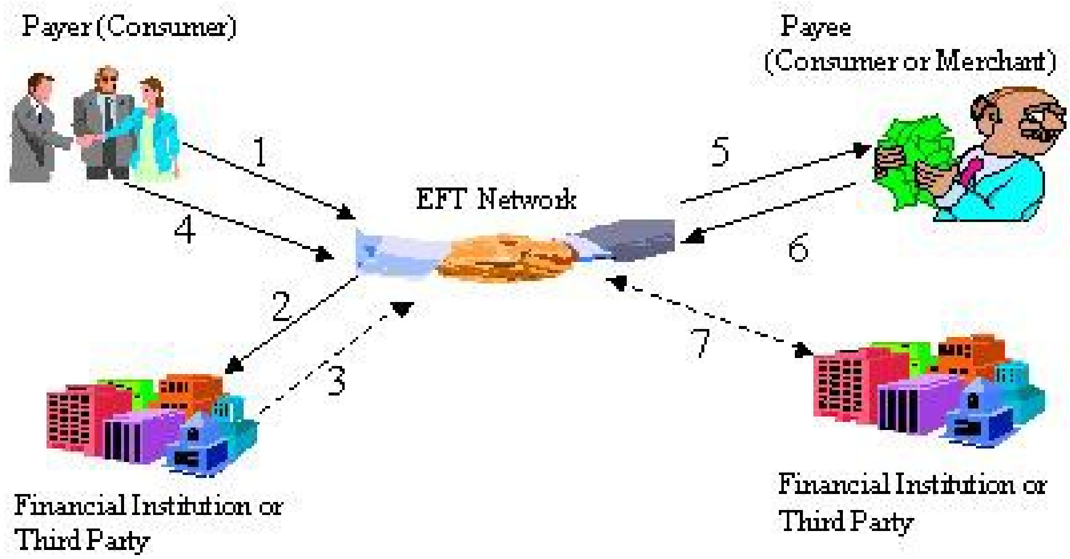
Most financial institution and nonbank ATM networks are connected to regional and national EFT/POS networks. Most regional EFT/POS networks are joint ventures owned and controlled by competing financial institutions, some function as cooperatives, and some are owned and operated by a single firm as a profit-making enterprise.

Visa and MasterCard own and operate the two national EFT/POS networks: (1) Visa's Plus and MasterCard's Cirrus ATM networks, and (2) Visa's Interlink and MasterCard's Maestro POS networks. The national networks serve as a bridge between regional networks, allowing them to route transaction information among them.

Membership in regional and national EFT/POS networks facilitates universal access to financial institution card-based electronic services and provides participant financial institutions with an interchange system offering authorization, clearing, and settlement services. Acquirers collect interchange fees from network members (issuers) to cover operating costs. With ATM transactions, the issuer pays fees to the acquirer, in contrast to credit and debit card networks in which the acquirer pays fees to the issuer.

Many financial institutions often rely on third-party service providers to conduct ATM and debit card payment processing. Third-party service providers provide a range of retail payment-related services, including card issuing, merchant, account maintenance and authorization, transaction routing and gateway, off-line debit processing, and clearing and settlement services. Although merchant acquiring financial institutions may use third-party service providers to perform many acquiring activities, the acquiring financial institution remains responsible for all third-party service-provider merchant activities.

Independent sales organizations (ISOs) provide third-party services to install and operate ATM and POS terminals for financial institutions and merchants. Representing merchants and community financial institutions, an ISO typically contracts with third-party service providers for a variety of services including support of ATM and POS terminals, transaction processing, and cash restocking. Some EFT/POS networks require an ISO to be sponsored by a financial institution member of the network.



Legend: Solid lines represent the flow of information and dashed lines represent the flow of funds.

Figure 6: PIN-based Debit Clearing and Settlement

Figure 6 describes a generic, online, PIN-based, debit card transaction. The consumer enters a PIN to authorize the transaction (Step 1). The merchant's financial institution requests authorization from the consumer's financial institution through the EFT/POS network (Step 2 and Step 3). The consumer's financial institution, or in some cases the regional network, verifies availability of funds and debits the consumer's account (step 4). The EFT/POS network contacts the merchant and authorizes the purchase (Step 5).

Typically, the acquiring financial institution does not credit the merchants' account with the entire amount of the transaction (similar to credit card clearing). Rather, the merchant receives the transaction amount, net of applicable fees and other expenses assessed by the acquiring financial institution and other intermediaries to the transaction (Step 6). For settlement, at the end of the business day, the regional EFT/POS networks determine the net debit and credit positions of the participating financial institutions and settle their positions using the ACH (Step 7).

Prepaid (Stored Value) Cards

The market for prepaid cards, sometimes called stored value cards, is one of the fastest growing segments of the retail financial services industry. While the terms prepaid cards and stored-value cards are frequently used interchangeably, differences exist between the two products. Prepaid cards are generally issued to persons who deposit funds into an account of the issuer. During the funds deposit process, most issuers establish an account and obtain identifying data from the purchaser (e.g., name, phone number, and etc.). Stored-value cards do not typically involve a deposit of funds as the value is prepaid and stored directly on the cards. Because its business model requires cardholders to pay in advance, it substantially eliminates the nonpayment risk for the issuing financial institution. The functionality of this product is leading to a wide range of card programs that operate in either closed or open-loop systems, and program innovation has resulted in the development of systems that operate in both structures. Closed-loop systems are generally retailer/issuer business models, while general-purpose cards issued by financial institutions tend to operate in open-loop systems. Open-loop system prepaid cards are processed using the same systems as the branded network cards - MasterCard, Visa, American Express, and Discover - and offer the same functionality.

In the past, prepaid cards were mostly issued by nonfinancial businesses in limited deployment environments such as mass transit systems and universities. In recent years, prepaid cards have grown significantly as financial institutions and nonbank organizations target under-banked markets and overseas remittances. Technological innovations in the way information is stored (e.g., magnetic strip or computer chip), the physical form of the payment mechanism, and biometric account access and authentication are converging to create efficiencies, reduce transaction times at the POS, and lower transaction costs.

There are several types of prepaid cards, including gift, payroll, travel, and teen cards. Either the consumer or an issuer funds the account for the card. When a consumer uses the card to make a purchase, the merchant deducts the amount of the purchase from the card. Transaction authorization can take place through an existing network, a chip stored on the card, or information coded on the magnetic strip. Once the stored value in the card is exhausted, customers may either replenish the value or acquire a new card.

In addition to cards, stored-value payment devices are emerging in a variety of other physical forms, most notably key fobs. With the recent introduction of contactless payment technologies, use of chips (smart cards), radio frequency identification (RFID), and near field communication (NFC) payment devices are becoming more innovative. Initiatives are underway to introduce mobile phones with integrated microchips that can initiate a payment when waved over a specially-equipped reader. The integrated chip can store value, authenticate a consumer, or contain consumer preferences and loyalty program information that can be used for marketing purposes.

Prepaid cards may be subject to legal and regulatory risks. For example, the Federal Reserve Board's final rule on Regulation E, issued August 30, 2006, extended its applicability to prepaid cards used for consumer's payroll. The Federal Reserve Board noted that it will monitor the development of other card products and may reconsider Regulation E coverage as these products continue to develop. State laws vary widely with regard to fees. Additionally, financial institutions should ensure that prepaid card product programs comply with the BSA and anti-money laundering guidance.

Payroll Cards

Payroll cards provide a means for paying a consumer's wages or other compensation in an access device with the functionality of a debit card. The card is loaded with the customer's payroll information on a magnetic strip or microchip and can be used to access an account that the employer establishes with a financial institution. The employee can use the payroll card to withdraw the funds at an ATM and to make POS purchases without a banking relationship. Some payroll cards may offer features such as convenience checks and electronic bill payment. Payroll cards are often marketed to employers as a cost-effective means of providing wages to employees who lack a traditional banking relationship. Their low-cost structure and debit-like functionality make them attractive as an alternative to direct deposit to more transient consumers. The Federal Reserve Board has amended its Regulation E to apply to payroll cards.

Payroll cards are supported by the Visa and MasterCard networks and can be used in every way that other branded cards are used. Employers are increasingly adopting payroll cards, and the growth is expected to continue because of their cost advantage to employers and financial institutions. Third-party service providers have sought opportunities in this market and may be engaged for card issuance, processing transactions made on the payroll card account, providing a range of program administration services for financial institutions or employers, and offering customer services to cardholders. Figure 7 illustrates the various relationships in an open-system payroll card program.

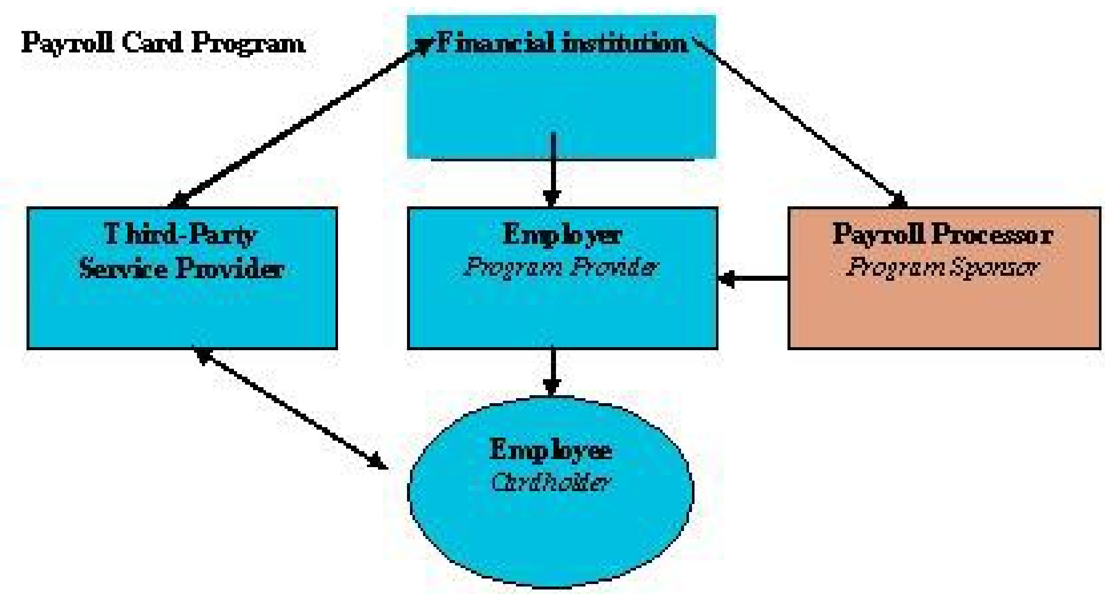


Figure 7: Open-system payroll card program

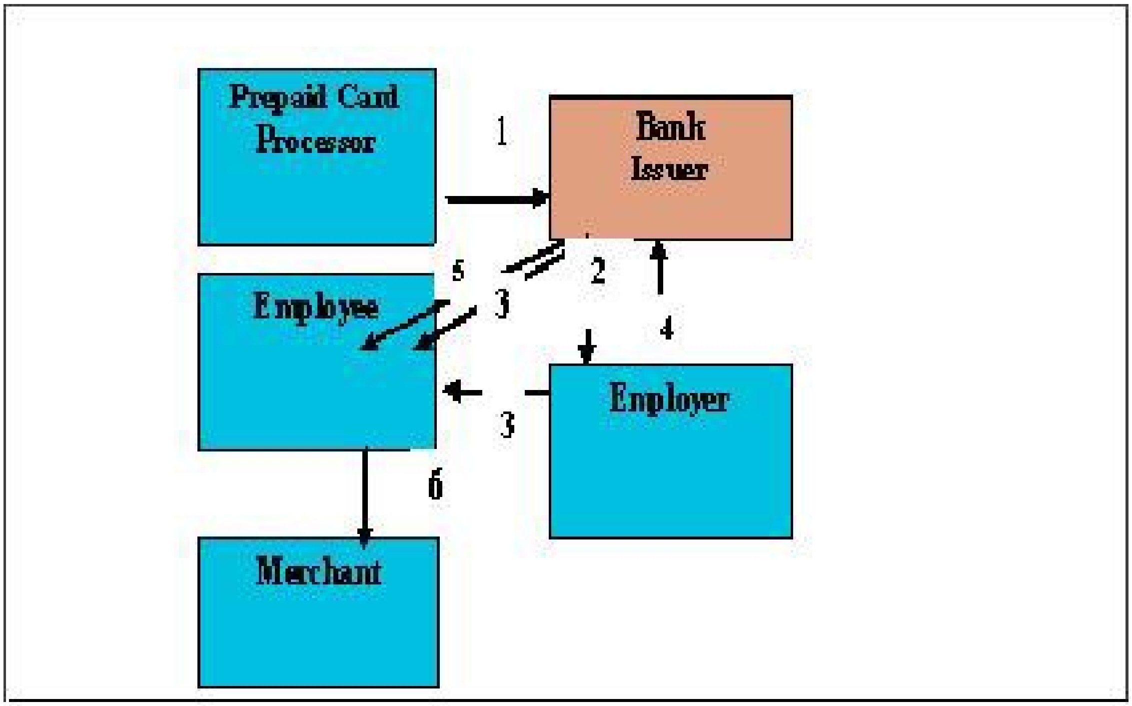


Figure 8: Stored value card product designed for corporate payrolls

Figure 8 describes a stored value card used in a payroll program. A stored value processor works with a financial institution to establish a payroll card program (Step 1). The issuer (financial institution) manages the card issuance and transaction processing. The financial institution offers the payroll card services to employers (Step 2). Either the financial institution or the employer distributes the payroll cards to employees (Step 3). The employer tells the financial institution the amount to credit to each employee's payroll card account (Step 4). On the pay date, the financial institution posts the funds to the employees' accounts (Step 5), allowing them to make purchases at any merchant that accepts the card's branding, e.g., Visa, MasterCard (Step 6).

General Spending Reloadable Cards

General spending card programs are offered by both financial institution and nonbank program providers or sponsors and are typically targeted to a particular consumer segment. Nonbank program providers usually sell this type of card and may have a relationship with a money service business or retailer, who, in turn, acts as agent for a nonbank program provider. See Figure 9 for a typical structure. Check-cashing businesses and convenience stores are examples of agents used by nonbank program providers. All network-branded prepaid cards must be issued by a partnering financial institution that is a member of the Visa or MasterCard networks or by American Express or Discover. There is a growing group of market participants associated with these programs and a developing range of potential functionality.